

"OR" +
"AND" .

COMP 2121 DISCRETE MATHEMATICS

LAB 1

1. How many 12 bit strings

a) start with 101?

1 0 1 0 1 0 1 (2) (2) (2) (2)

$$2^9$$

b) start with 011 or with 0000?

0 1 1 0 0 0 0 2^9 2^8

$$2^9 + 2^8$$

2. How many hexadecimal strings of length 8

a) do not contain letter A?

15 15 15 15^8

b) contain at least one letter A?

≥ 1

A A A A A A A A 16^7 16^7 16^7 16^7

$8(16)^7$ is NOT correct! overcounting!

16^8

A	A	A	A	A	A	A	A	A
0	1	2	3	4	5	6	7	8

same!

$$16^8 - 15^8$$

$$2) \frac{8!}{1!7!} 15^7 + \frac{8!}{2!6!} 15^6 + \frac{8!}{3!5!} 15^5 + \frac{8!}{4!4!} 15^4 + \frac{8!}{5!3!} 15^3 + \frac{8!}{6!2!} 15^2 + \frac{8!}{7!1!} 15 + 1$$

c) contain exactly one letter A?

d) contain at most one letter A?

≤ 1

8 { A A A A A A A A 15^7 15^7 15^7 15^7

$$15^8 + 8(15)^7$$

$$8(15)^7$$

3. What value does a variable *counter* have after the following code execution?

<pre> counter = 100 for i = 0 to 10 { counter = counter + 2 } for j = 1 to 20 { counter = counter + 3 for k = 2 to 30 { counter = counter + 4 } } </pre>	1
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$$i: 10 - 0 + 1 = 11$$

$$j: 20$$

$$k: 30 - 2 + 1 = 29$$

$$\text{counter} = 100 + \frac{11}{i} (2) + \frac{20}{j} (3) + \frac{20(29)}{j,k} (4) = 2502$$

<pre> counter = a for i = 1 to n { counter = counter + 2 for j = n to 2n { counter = counter + 3 for k = 0 to 3n { counter = counter + 4 } } } // assume n ≥ 1 </pre>	2
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$$i \sim n$$

$$j \sim 2n - n + 1 = n + 1$$

$$k \sim 3n - 0 + 1 = 3n + 1$$

$$\text{counter} = a + \frac{n}{i} (2) + \frac{n(n+1)}{i,j} (3) + \frac{n(n+1)(3n+1)}{i,j,k} (4)$$

$$= a + 2n + (n^2 + n)(3) + (n^2 + n)(12n + 4)$$

$$= a + 2n + 3n^2 + 3n + 12n^3 + 4n^2 + 12n^2 + 4n$$

$$= 12n^3 + 19n^2 + 9n + a$$

4. Consider passwords with the following restrictions:

✓ Length 12.

✓ Available characters are 26 upper case letters A, B, C... Z.

a) What is the total number of passwords?

$26 \quad 26 \quad \dots \quad 26$
 26^{12}

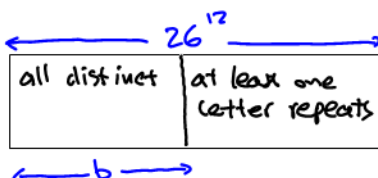
b) How many passwords have all distinct letters?

$26 \quad 25 \quad \dots \quad 15$

$P(26, 12) = \frac{26!}{(26-12)!}$

$26(25)(24)(23)(22)(21)(20)(19)(18)(17)(16)(15) = \frac{26!}{14!}$

c) How many passwords have at least one letter repeated?



$26^{12} - \frac{26!}{14!}$

d) How many passwords have four letters "A", six letters "B" and two letters "C"?

AAAA BB BB BB CC

"MISSISSIPPI PROBLEM"

$\frac{12!}{4! 6! 2!}$

e) How many passwords have seven letters "X" and four letters "Y" (and one other letter)? *neither X nor Y*

$\frac{12!}{7! 4!}$
 $\left\{ \begin{array}{ll} \text{XXXX XXXY YYY-} & (24) \\ \text{XXXX YYYYY X-XX} & (24) \\ \vdots & \\ \text{YXXXX Y-YY XXXX} & (24) \end{array} \right.$

$\frac{12!}{7! 4!} (24)$

5. Consider the string COMPUTERARCHITECTURE

a) How many arrangements are there of all the letters?

$$\frac{20!}{3!2!3!3!3!}$$

b) How many arrangements in part a) start with HA?

HA THE REST

$$\frac{18!}{3!2!3!3!3!}$$

c) How many arrangements in part a) do not start with HA?

$$\frac{20!}{3!2!3!3!3!} - \frac{18!}{3!2!3!3!3!}$$

d) How many arrangements in part a) have all vowels together?

OUUEEEAI CCCMP TTT RRR H

CCCMP VOWELS TTT RRR H

$$\frac{13!}{3!3!3!} \cdot \frac{8!}{2!3!}$$

e) How many arrangements in part a) have all yowels together and all consonants together?

$$\frac{8!}{2!3!} \cdot \frac{12!}{3!3!3!} \cdot 2$$

VC or CV

CLC
O
M
P
UU
TTT
EEE
RRR
A X
H X
I

- a) How many different messages are there?

28 28 28

28¹⁰

- $\overline{2828} \quad \overline{19}$

$$P(28, 10) = \frac{28!}{18!}$$

- i. 10 #s

i. 10 #s

one

$28 - \# = 27$

ii. 9 #s

$(10)(27) = 270$

- iii. 4 #s

4 #s

$$\frac{10!}{4!6!}$$

{

## ## # -	27	27	27 ⁶
## ## - #	27	27	27 ⁶
# - # - # #	27	27	27 ⁶
⋮			
- - - - # ## ##	27	27	27 ⁶

$$\frac{10!}{4!6!} (27)^6$$

d) In how many different ways can the message CST be transmitted?

CST $\longrightarrow$

CST	22	222222
2C2S2	22	T22
:		
222C2	S2	222T

 $\longrightarrow$ CST

[illegible]

If all of the problems can't be done in the lab period, students can work on them on their own.